

ML2 Dataset Writeup: Group K

Members:

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Task (regression):

Predicting the next season batting average of arbitrary baseball players given the previous year's statistics.

ML Methods:

1. Gradient Boosted Decision Tree (ML2)
2. Regularized Regression - Lasso and Ridge (ML1)
3. TabPFN*

Dataset:

Standard and advanced batting statistics in the 2020-2024 baseball seasons.

Dimensions:

(1455, 41) including target, season, and 2 ID columns (37 features)

Source:

<https://www.fangraphs.com/leaders/> (Copy pasted table manually)

For Splits: (Test -> 1. Split on players, 2. Use the most current year)

We plan to drop the player_id and year_id when doing machine learning to make the model more generalizable to arbitrary players and arbitrary years.

Data Summary (Disclaimer, AI was used to generate the table):

* TabPFN will be included if time permits in the appendix. The primary models to be assessed are the Gradient Boosted Decision Tree and the Regularized Regression.

Column Name	Description	Data Type
Season	Year of the season	int
Name	Player full name	str
Team	Team abbreviation	str
G	Games played	int
AB	At-bats	int
H	Hits	int
X1B	Singles	int
X2B	Doubles	int
X3B	Triples	int
HR	Home runs	int
R	Runs scored	int
RBI	Runs batted in	int
BB.Percent	Walk rate (decimal, e.g., 0.057 = 5.7%)	float
IBB	Intentional walks	int
SO	Strikeouts	int
HBP	Hit by pitch	int
SF	Sacrifice flies	int
SH	Sacrifice hits / bunts	int
GDP	Grounded into double play	int
SB	Stolen bases	int
CS	Caught stealing	int
PA	Plate appearances	int

K.Percent	Strikeout rate (decimal, e.g., 0.214 = 21.4%)	float
BB.K	Walk-to-strikeout ratio	float
AVG	Batting average	float
OBP	On-base percentage	float
SLG	Slugging percentage	float
OPS	On-base plus slugging	float
ISO	Isolated power	float
Spd	Speed score	float
BABIP	Batting average on balls in play	float
UBR	Ultimate base running	float
wGDP	Weighted GDP	float
XBR	Expected runs	float
wSB	Weighted stolen bases	float
wRC	Weighted runs created	float
wRAA	Weighted runs above average	float
wOBA	Weighted on-base average	float
wRC.	Weighted runs created (normalized)	float
next_AVG	Next season's batting average (target)	float